RoySearch: Model Notes

Two-Ability Comparative Advantage in Segmented Search Markets

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Scope. This document describes the model: its environment, the values and decisions of workers and firms, the stationary equilibrium, the transition between two stationary equilibria, the model-implied Beveridge curve, and the algorithms that solve them. How the model is taken to the data, from the moments and their construction to identification and the status of every parameter, is in the companion *Data and Moments* notes.

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I build a model of training and employment in two frictional labour markets, with workers who differ in a pair of aptitudes. Each market draws from Mortensen and Pissarides (1994) as far as match creation and destruction are concerned. In the skilled market match quality is drawn on contact, workers search on the job, and the ladder they climb is one in match quality, following Pissarides (2000). Wages are bargained in both markets. The training choice is a Roy (1951) problem, and the dependence between the two aptitudes is modelled as in Gola (2021). Wages change only with match quality: the model abstracts from skill accumulation on the job.

Every value a worker holds depends on a single aptitude, except through the two choices that compare them: whether to train, and which market a skilled unemployed worker searches. Three assumptions deliver this: output in each market depends only on the aptitude for that market, a skilled unemployed worker searches one market at a time, and only workers in skilled matches search on the job, in the skilled market.

1 Workers, jobs and matches

I focus on steady states, and the transition between two of them. Time is continuous, and all agents discount at rate $r > 0$.

1.1 Workers

Each worker is characterised by an innate, permanent pair of abilities $\mathbf{a} = (a_U, a_S) \in [0,1]^2$, fixed over time, where a_U is aptitude for unskilled work and a_S is aptitude for skilled work. Abilities do not change with employment, training, or age; training changes *market access only*, not $\mathbf{a}$.

Marginals. Each ability is its rank in the population: a_U and a_S are each uniformly distributed.

Copula. The joint distribution is a Gaussian copula with a single correlation $\rho_x \in (-1,1)$ on the Gaussian ranks:

$$L(a_U, a_S) = \Phi_{\rho_x}\big(\Phi^{-1}(a_U),\, \Phi^{-1}(a_S)\big),$$

where Φ_{ρ_x} is the bivariate standard normal CDF with correlation ρ_x. Because the marginals are uniform, the joint density is the copula density itself,

$$\ell(a_U, a_S) = c_{\rho_x}(a_U, a_S), \qquad c_{\rho_x}(u,v) = \frac{\exp\Big(-\frac{\rho_x^2(\zeta_u^2+\zeta_v^2)-2\rho_x\zeta_u\zeta_v}{2(1-\rho_x^2)}\Big)}{\sqrt{1-\rho_x^2}}, \tag{1}$$

with $\zeta_u = \Phi^{-1}(u)$, $\zeta_v = \Phi^{-1}(v)$. Newborns are drawn from $\ell(a_U, a_S)$.

Demographic turnover. Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn unskilled workers drawn from the population type distribution $\ell(a_U, a_S)$. This delivers a stationary population composition. The rate ν is the model's only always-on renewal clock; it fires identically in every stationary equilibrium and along every transition path.

1.2 Training and market access

There are two labour markets, unskilled U and skilled S. A worker is born unskilled and unemployed. As an unskilled worker she can work only in the unskilled market, and the way into the skilled market is training, which she can begin only while unemployed. An unskilled unemployed worker therefore chooses between searching for an unskilled job ($\tau(a_U, a_S) = 0$) and starting training ($\tau(a_U, a_S) = 1$). Training takes effort and time and develops no human capital: it only grants access to the skilled market. A worker who completes it is skilled and unemployed, and chooses which of the two markets to search. In the skilled market she may keep searching once employed, and she moves whenever she meets a better match; in the unskilled market workers do not search on the job. If a skilled worker accepts an unskilled job she loses her skilled status, and from then on she is an unskilled worker like any other. A match in either market ends when a productivity shock makes it unprofitable; the worker then returns to unemployment in her market, where the same choices start again. At any point a worker may leave the economy, and a newborn takes her place.

Cost. An unskilled unemployed worker can train to become skilled unemployed. Initiating training requires a one-time psychic cost of effort, which is decreasing in *skilled* aptitude:

$$\mathcal{C}(a_S) = c\,(1 - a_S)\,e^{-a_S}, \qquad c > 0. \tag{2}$$

Completion. Training completes at Poisson rate $\phi > 0$; on completion the worker becomes skilled and enters skilled unemployment.

Search by the skilled unemployed. A skilled unemployed worker searches one market at a time: $d(a_U, a_S) = 1$ if she searches the unskilled market and $d(a_U, a_S) = 0$ if she searches the skilled market. She keeps her skilled status while she searches the unskilled market.

1.3 Jobs and production

Both markets have homogeneous firms. A firm in market j posts a vacancy at flow cost k_j; the output of a match depends only on the worker's aptitude for that market (a_U in U, a_S in S) and on match quality p. Output in both sectors is scaled by a general productivity level A and by sector-specific productivity shifters P_U and P_S. I normalise $P_U = 1$. Both outputs are linear and increasing in their respective aptitudes.

$$\pi_U(a_U, p) \;=\; A\,P_U\,a_U\,p, \qquad \pi_S(a_S, p) \;=\; A\,P_S\,a_S\,p. \tag{3}$$

1.4 Match quality and shocks

Match quality $p \in [0, 1]$ is specific to each match. A new unskilled match starts at $p = 1$, and a new skilled match draws its quality from the uniform distribution on $[0, 1]$. Quality shocks arrive at Poisson rate λ_j; when one arrives, a new quality p' is drawn from the distribution with cdf $G(p) = p^{\alpha_U}$ in the unskilled market and with cdf $\Gamma(p) = p$ in the skilled market. A higher α_U makes high new qualities more likely and raises the mean new quality, $\alpha_U/(1 + \alpha_U)$, so α_U measures the quality of unskilled redraws.

1.5 States

Equilibrium results in a distribution of workers across states. For a worker of type (a_U, a_S), let u_U, e_U, t and u_S denote the densities of unskilled unemployment, unskilled employment, training and skilled unemployment. Skilled employment is distributed over both worker type (a_U, a_S) and match quality p, since a skilled worker's search on the job depends on the quality of her current match: $e_S(a_U, a_S, p)$ is the density of skilled matches of quality p. The densities satisfy the following balance equation:

$$u_U(a_U, a_S) + e_U(a_U, a_S) + t(a_U, a_S) + u_S(a_U, a_S) + \int e_S(a_U, a_S, p)\,\mathrm{d}p \;=\; \ell(a_U, a_S). \tag{4}$$

2 Meetings, wages and match formation

2.1 Meetings

While unskilled unemployed workers can only search the unskilled market, a skilled unemployed worker can make a choice to search exclusively in one of the two markets: the skilled market if $d(a_U, a_S) = 0$ and the unskilled market if $d(a_U, a_S) = 1$. A skilled worker may also search on the job, in the skilled market only, at flow cost σ_S: OJS indicator $s(a_S, p) = 1$ if a worker with skilled aptitude a_S in a match of quality p searches on the job and zero otherwise.

The job seeker masses of the two markets are

$$\begin{aligned}
\widetilde{\mathcal{U}}_U &\equiv \iint \big(u_U(a_U, a_S) + d(a_U, a_S)\, u_S(a_U, a_S)\big)\,\mathrm{d}a_U\,\mathrm{d}a_S, \\
\widetilde{\mathcal{U}}_S &\equiv \iint \Big(\big(1 - d(a_U, a_S)\big)\, u_S(a_U, a_S) + \int s(a_S, p)\, e_S(a_U, a_S, p)\,\mathrm{d}p\Big)\,\mathrm{d}a_U\,\mathrm{d}a_S.
\end{aligned}$$

In each market $j \in \{U, S\}$, the number of meetings between vacancies $\mathcal{V}_j$ and job seekers $\widetilde{\mathcal{U}}_j$ is given by the matching function $M_j = \mu_j \sqrt{\widetilde{\mathcal{U}}_j\, \mathcal{V}_j}$, with $\mu_j > 0$. Tightness is $\theta_j = \mathcal{V}_j / \widetilde{\mathcal{U}}_j$. A job seeker in market j meets a vacancy at rate

$$\kappa_j \equiv \frac{M_j}{\widetilde{\mathcal{U}}_j} = \mu_j \sqrt{\theta_j},$$

whether she is unemployed or searching on the job, and a vacancy meets a job seeker at rate $M_j / \mathcal{V}_j = \kappa_j / \theta_j$.

2.2 Wages

In each market $j \in \{U, S\}$, let $E_j(a_j, p)$ and $J_j(a_j, p)$ denote the values to a worker and a firm of a match of quality p, where a_j is the worker's aptitude for market j; U_j the value of unemployment; and V_j the value of a vacancy. The surplus of a match is

$$S_j(a_j, p) \equiv E_j(a_j, p) - U_j + J_j(a_j, p) - V_j. \tag{5}$$

Wages are set by Nash bargaining with worker weight $\beta_j \in (0, 1)$, which splits the surplus as

$$E_j(a_j, p) - U_j = \beta_j\, S_j(a_j, p), \qquad J_j(a_j, p) - V_j = (1 - \beta_j)\, S_j(a_j, p). \tag{6}$$

The wage of a match depends on the worker's ability in that market, on the match quality and, in the skilled market, on whether the worker searches on the job. A wage is set when a match forms and is bargained again whenever the match quality changes. When a worker who searches on the job moves to a new match, her new wage is bargained against U_S, and the employer she leaves makes no counter-offer.

2.3 Match formation

Both gains in (6) take the sign of S_j, so a meeting becomes a match, and a match survives a productivity shock, if and only if $S_j(a_j, p) \geq 0$. A worker who searches on the job and meets a vacancy whose match quality is higher than that of her current match moves to it.

3 Value functions

The values of workers and jobs have been kept implicit up to now. They define each agent's decisions: whether to train, which market to search, whether to search on the job, and which matches to keep.

3.1 Unskilled unemployed workers

Let $U_U^{(0)}$ be the value of unskilled search and T the value of training. An unskilled unemployed worker of type (a_U, a_S) chooses between unskilled search and training:

$$U_U(a_U, a_S) = \max\{U_U^{(0)}(a_U),\ U_U^{(1)}(a_U, a_S)\}, \qquad U_U^{(1)}(a_U, a_S) = -\mathcal{C}(a_S) + T(a_U, a_S). \tag{7}$$

Training yields no flow payoff and completes at rate ϕ into skilled unemployment, and the completing worker collects the value of skilled unemployment, $U_S(a_U, a_S)$:

$$(r + \phi + \nu)\, T(a_U, a_S) = \phi\, U_S(a_U, a_S). \tag{8}$$

Unskilled search yields unemployment income b_U, and the worker meets a vacancy at rate κ_U:

$$(r + \nu)\, U_U^{(0)}(a_U) = b_U + \kappa_U \big[E_U(a_U, 1) - U_U^{(0)}(a_U)\big]^+, \tag{9}$$

where $[x]^+ = \max\{x, 0\}$: she takes the job if it raises her value. The worker trains, $\tau(a_U, a_S) = 1$, if and only if $U_U^{(1)}(a_U, a_S) \geq U_U^{(0)}(a_U)$.

3.2 Unskilled matches

An unskilled match of quality p produces $AP_U a_U\, p$; the worker receives the wage $w_U(a_U, p)$ and the firm keeps the rest. A productivity shock arrives at rate λ_U and draws a new quality p' from the distribution with cdf G. The match continues at the new quality if its surplus there is nonnegative, which is the case when p' is at least a reservation quality $p^*(a_U)$, characterised below. Otherwise it ends: the worker returns to unskilled unemployment, worth U_U, and the firm is left with nothing. The match also ends when the worker leaves the economy, at rate ν. The values of the match to the worker and to the firm satisfy

$$(r + \nu + \lambda_U)\, E_U(a_U, p) = w_U(a_U, p) + \lambda_U \Big[G(p^*(a_U))\, U_U + \int_{p^*(a_U)}^1 E_U(a_U, p')\, \mathrm{d}G(p')\Big], \tag{10}$$

$$(r + \nu + \lambda_U)\, J_U(a_U, p) = A\, P_U\, a_U\, p - w_U(a_U, p) + \lambda_U \int_{p^*(a_U)}^1 J_U(a_U, p')\, \mathrm{d}G(p'). \tag{11}$$

Adding the two and subtracting the worker's unemployment value gives the surplus (5) of the match,

$$(r+\nu+\lambda_U)\,S_U(a_U,p) = A\,P_U\,a_U\,p - (r+\nu)\,U_U + \lambda_U \int_{p^*(a_U)}^{1} S_U(a_U,p')\,\mathrm{d}G(p'). \tag{12}$$

The wage does not enter it: the surplus, and with it the reservation quality, is set by output, the worker's unemployment value and the shock process.

Wage. The sharing rule (6) and (10)–(11) give the wage

$$w_U(a_U,p) = \beta_U\,A\,P_U\,a_U\,p + (1-\beta_U)(r+\nu)\,U_U. \tag{13}$$

Separation. The surplus increases in p, so a match survives exactly when its quality is at least the reservation quality $p^*(a_U)$, defined by

$$p^*(a_U) = \min\{p : S_U(a_U,p) \geq 0\}. \tag{14}$$

Where $p^*(a_U)$ is interior, $S_U\big(a_U,p^*(a_U)\big) = 0$, and evaluating (12) there gives

$$p^*(a_U) = \frac{(r+\nu)\,U_U - \lambda_U \int_{p^*(a_U)}^{1} S_U(a_U,p')\,\mathrm{d}G(p')}{AP_U\,a_U}. \tag{15}$$

3.3 Skilled unemployed workers

Let $U_S^{(0)}$ be the value of searching the skilled market and $U_S^{(1)}$ the value of searching the unskilled market. A skilled unemployed worker of type (a_U,a_S) chooses between the two:

$$U_S(a_U,a_S) = \max\{U_S^{(0)}(a_S),\ U_S^{(1)}(a_U)\}. \tag{16}$$

Skilled search yields unemployment income b_S, and the worker meets a vacancy at rate κ_S:

$$(r+\nu)\,U_S^{(0)}(a_S) = b_S + \kappa_S \int \big[E_S(a_S,p) - U_S^{(0)}(a_S)\big]^+\,\mathrm{d}\Gamma(p). \tag{17}$$

Search in the unskilled market yields the same unemployment income b_S, and the worker meets a vacancy at rate κ_U:

$$(r+\nu)\,U_S^{(1)}(a_U) = b_S + \kappa_U\big[E_U(a_U,1) - U_S^{(1)}(a_U)\big]^+. \tag{18}$$

The worker searches the unskilled market, $d(a_U,a_S)=1$, if and only if $U_S^{(1)}(a_U) > U_S^{(0)}(a_S)$. In the values of skilled matches below, U_S is $U_S(a_U,a_S)$ of (16).

3.4 Skilled matches

Let $E_S^{(0)}$ and $J_S^{(0)}$ denote the values of a skilled match to the worker and the firm when the worker does not search on the job, and $E_S^{(1)}$ and $J_S^{(1)}$ when she does, so that $E_S = \max\{E_S^{(0)},E_S^{(1)}\}$, and let J_S denote the firm's value in the branch the worker chooses.

A skilled match of quality p produces $AP_S a_S\,p$; the worker receives the wage $w_S^{(0)}(a_S,p)$ if she does not search on the job and $w_S^{(1)}(a_S,p)$ if she does, and the firm keeps the rest. A productivity

shock arrives at rate λ_S and draws a new quality p' from the distribution with cdf Γ. The match continues at the new quality if p' is at least a reservation quality $p_S^*(a_S)$, characterised below, and otherwise ends. It also ends at rate ν when the worker leaves the economy. When it ends, the worker returns to skilled unemployment, worth U_S, and the firm is left with nothing. A worker who searches on the job pays the flow cost σ_S and meets a vacancy at rate κ_S; the quality $\tilde{p}$ of the new match is drawn from the distribution with cdf Γ, and she moves if $\tilde{p} > p$, leaving her firm with nothing. The values of the match satisfy

$$\begin{aligned}(r+\nu+\lambda_S)\,E_S^{(0)}(a_S,p) &= w_S^{(0)}(a_S,p) \\ &\quad + \lambda_S\Big[\Gamma(p_S^*)\,U_S + \int_{p_S^*}^1 E_S(a_S,p')\,\mathrm{d}\Gamma(p')\Big], \qquad (19)\\ \big(r+\nu+\lambda_S+\kappa_S(1-\Gamma(p))\big)\,E_S^{(1)}(a_S,p) &= w_S^{(1)}(a_S,p) - \sigma_S \\ &\quad + \lambda_S\Big[\Gamma(p_S^*)\,U_S + \int_{p_S^*}^1 E_S(a_S,p')\,\mathrm{d}\Gamma(p')\Big] \\ &\quad + \kappa_S\int_p^1 E_S(a_S,\tilde{p})\,\mathrm{d}\Gamma(\tilde{p}), \qquad (20)\\ (r+\nu+\lambda_S)\,J_S^{(0)}(a_S,p) &= AP_Sa_Sp - w_S^{(0)}(a_S,p) + \lambda_S\int_{p_S^*}^1 J_S(a_S,p')\,\mathrm{d}\Gamma(p'), \qquad (21)\\ \big(r+\nu+\lambda_S+\kappa_S(1-\Gamma(p))\big)\,J_S^{(1)}(a_S,p) &= AP_Sa_Sp - w_S^{(1)}(a_S,p) \\ &\quad + \lambda_S\int_{p_S^*}^1 J_S(a_S,p')\,\mathrm{d}\Gamma(p'). \qquad (22)\end{aligned}$$

Adding the worker's and the firm's values in each branch and subtracting the worker's unemployment value gives the surpluses $S_S^{(k)} = E_S^{(k)} - U_S + J_S^{(k)}$, $k \in \{0,1\}$:

$$\begin{aligned}(r+\nu+\lambda_S)\,S_S^{(0)}(a_S,p) &= AP_Sa_Sp - (r+\nu)U_S + \lambda_S\int_{p_S^*}^1 S_S(a_S,p')\,\mathrm{d}\Gamma(p'), \qquad (23)\\ \big(r+\nu+\lambda_S+\kappa_S(1-\Gamma(p))\big)\,S_S^{(1)}(a_S,p) &= AP_Sa_Sp - (r+\nu)U_S - \sigma_S \\ &\quad + \lambda_S\int_{p_S^*}^1 S_S(a_S,p')\,\mathrm{d}\Gamma(p') \\ &\quad + \kappa_S\beta_S\int_p^1 S_S(a_S,\tilde{p})\,\mathrm{d}\Gamma(\tilde{p}), \qquad (24)\end{aligned}$$

where the last term uses $E_S(a_S,\tilde{p}) - U_S = \beta_S S_S(a_S,\tilde{p})$: a worker who moves takes the share β_S of the surplus of her new match.

On-the-job search. By the sharing rule (6), $E_S^{(k)} - U_S = \beta_S S_S^{(k)}$, so the worker searches on the job, $s(a_S,p) = 1$, if and only if $S_S^{(1)}(a_S,p) \geq S_S^{(0)}(a_S,p)$, and $S_S = \max\{S_S^{(0)}, S_S^{(1)}\}$. Let $p_S^{\mathrm{oj}}(a_S)$ denote the highest quality at which she searches.

Wages. The sharing rule (6) and (19)–(22) give the wage of a worker who does not search on the job,

$$w_S^{(0)}(a_S, p) = \beta_S A P_S a_S p + (1 - \beta_S)(r + \nu)U_S, \tag{25}$$

and of one who does,

$$w_S^{(1)}(a_S, p) = \beta_S A P_S a_S p + (1-\beta_S)(r+\nu)U_S + (1-\beta_S)\sigma_S - \beta_S(1-\beta_S)\kappa_S \int_p^1 S_S(a_S, \tilde{p})\,\mathrm{d}\Gamma(\tilde{p}). \tag{26}$$

Separation. The surplus is increasing in p wherever it is nonnegative, so a match survives exactly when its quality is at least the reservation quality $p_S^*(a_S)$, defined by

$$p_S^*(a_S) = \min\{p : S_S(a_S, p) \geq 0\}. \tag{27}$$

3.5 Vacant jobs

Let $\mathcal{J}_j$ denote the expected value to the firm of a meeting in market j. A vacancy in market j costs its firm the flow k_j and meets a job seeker at rate κ_j/θ_j. In the unskilled market the job seeker it meets is either an unskilled unemployed worker or a skilled unemployed worker who searches the unskilled market. In both cases the match starts at quality 1 if the firm gains from it.

$$\mathcal{J}_U = \frac{1}{\widetilde{\mathcal{U}}_U} \iint \big[J_U(a_U, 1) - V_U\big]^+ \big(u_U(a_U, a_S) + d(a_U, a_S)\, u_S(a_U, a_S)\big)\,\mathrm{d}a_U\,\mathrm{d}a_S$$

The value of an unskilled vacancy satisfies

$$rV_U = -k_U + \frac{\kappa_U}{\theta_U}\,\mathcal{J}_U. \tag{28}$$

In the skilled market the job seeker it meets is either a skilled unemployed worker who searches the skilled market or a worker employed in a skilled match of quality p who searches on the job. A match with an unemployed worker draws its quality p from the distribution with cdf Γ; an employed worker draws $\tilde{p}$ from the same distribution and moves only if $\tilde{p} > p$.

$$\mathcal{J}_S = \frac{1}{\widetilde{\mathcal{U}}_S} \iint \Big[\big(1 - d(a_U, a_S)\big)\, u_S(a_U, a_S) \int \big[J_S(a_S, p) - V_S\big]^+ \mathrm{d}\Gamma(p) + \int s(a_S, p)\, e_S(a_U, a_S, p) \int_p^1 \big[J_S(a_S, \tilde{p}) - V_S\big]^+ \mathrm{d}\Gamma(\tilde{p})\,\mathrm{d}p\Big]\,\mathrm{d}a_U\,\mathrm{d}a_S$$

The value of a skilled vacancy satisfies

$$rV_S = -k_S + \frac{\kappa_S}{\theta_S}\,\mathcal{J}_S. \tag{29}$$

4 Steady-state equilibrium

In a steady state, tightness and every value, decision and density are constant.

4.1 Steady-state flows

In a stationary equilibrium the flows into and out of every state balance for every type. Every density and decision below is a function of the type (a_U, a_S), and I drop the arguments. Newborns and workers whose unskilled match ends enter unskilled unemployment, and those of a type that trains start training at once. Trainees leave by completing, at rate ϕ, or by leaving the economy, at rate ν:

$$\tau \left[\nu\, \ell + \lambda_U G(p^*)\, e_U\right] = (\phi + \nu)\, t. \tag{30}$$

The remaining unskilled unemployed leave by finding a job or by leaving the economy:

$$(1-\tau) \left[\nu\, \ell + \lambda_U G(p^*)\, e_U\right] = \left(\kappa_U \mathbf{1}\{S_U(\cdot, 1) \geq 0\} + \nu\right) u_U. \tag{31}$$

Unskilled matches form when unskilled or skilled unemployed searchers meet a feasible unskilled match, and end when a shock redraws their quality below p^* or when the worker leaves:

$$\kappa_U \mathbf{1}\{S_U(\cdot, 1) \geq 0\} \left(u_U + d\, u_S\right) = \left(\lambda_U G(p^*) + \nu\right) e_U. \tag{32}$$

Skilled unemployed workers arrive when trainees complete or a shock redraws the quality of a skilled match below p_S^*, and leave when they find a feasible match in the market they search or leave the economy:

$$\phi\, t + \lambda_S \Gamma(p_S^*) \int e_S(\cdot, p)\, \mathrm{d}p = \Big((1-d)\, \kappa_S \big(1 - \Gamma(p_S^*)\big) + d\, \kappa_U \mathbf{1}\{S_U(\cdot, 1) \geq 0\} + \nu\Big)\, u_S. \tag{33}$$

Skilled matches of quality p form when skilled unemployed searchers find a match, when a shock redraws a match to p, and when workers who search on the job move up. They end when a shock redraws their quality, when the worker leaves or when she moves up:

$$(1-d)\, \kappa_S\, u_S + \lambda_S \int e_S(\cdot, q)\, \mathrm{d}q + \kappa_S \int_{p_S^*}^{p} s(\cdot, q)\, e_S(\cdot, q)\, \mathrm{d}q = \big(\lambda_S + \nu + s\, \kappa_S (1 - \Gamma(p))\big)\, e_S(\cdot, p) \tag{34}$$

for $p \geq p_S^*$, and $e_S = 0$ below p_S^*. Adding the flow conditions of a type over its states gives (4).

4.2 Free entry

Firms open vacancies until a vacancy is worth nothing, $V_U = V_S = 0$. Free entry then requires that in market j, a vacancy's flow cost equal the rate at which it meets a job seeker times the expected value of a meeting:

$$\frac{\kappa_j}{\theta_j}\, \mathcal{J}_j = k_j. \tag{35}$$

The ratio κ_j/θ_j is the rate at which a vacancy meets job seekers. A tighter market always gives each vacancy fewer meetings. The expected value of a meeting $\mathcal{J}_j$ can rise with tightness if a tighter market adds to the pool job seekers who are better meetings than those already searching. This can come from search on the job, from workers who start to lose their matches, or from the choices of whether to train and which market to search. If $\mathcal{J}_j$ rises by enough to offset the fall in meetings, free entry may hold at the same cost at more than one tightness.

4.3 Stationary equilibrium

A stationary equilibrium is a tightness pair (θ_U, θ_S), reservation qualities p^* and p_S^*, decisions τ, d and s, wages w_U and w_S, and densities that satisfy the free-entry conditions (35), the job destruction conditions (14) and (27), the decision rules of Section 3, the wage equations (13), (25) and (26), and the steady-state flow conditions of Section 4.1.

At given vacancy costs, free entry can hold at more than one tightness pair, so the model can have several stationary equilibria at the same parameters. Given a tightness pair, however, free entry holds at a unique pair of vacancy costs.

Proposition 1 (Vacancy costs given tightness)**.** *At every tightness pair (θ_U, θ_S), free entry holds at exactly one pair of vacancy costs, $k_j = (\kappa_j/\theta_j)\,\mathcal{J}_j$ for $j \in \{U, S\}$.*

Proof. The vacancy costs enter the model only through free entry. They appear in no value, wage, reservation quality, decision or density. The ratio $\kappa_j/\theta_j = \mu_j \theta_j^{-1/2}$ depends on θ_j alone, and $\mathcal{J}_j$ is an integral of the firms' values against the job seekers' densities, both of which are uniquely determined by the tightness pair. The tightness pair therefore uniquely determines vacancy costs k_j in the free entry equations (35). □

The tightness pair can thus take the place of the vacancy costs among the model's unknowns: each tightness pair then corresponds to exactly one stationary equilibrium, together with the vacancy costs that support it.

5 The model-implied Beveridge curve

The labour force of market j consists of the unemployed workers who search it and the workers in its matches. Let $\bar{u}_j$ be the share of the labour force of market j that is unemployed:

$$\bar{u}_U = \frac{\iint (u_U + d\,u_S)\,\mathrm{d}a_U\,\mathrm{d}a_S}{\iint (u_U + d\,u_S + e_U)\,\mathrm{d}a_U\,\mathrm{d}a_S}, \qquad \bar{u}_S = \frac{\iint (1-d)\,u_S\,\mathrm{d}a_U\,\mathrm{d}a_S}{\iint \big((1-d)\,u_S + \int e_S\,\mathrm{d}p\big)\,\mathrm{d}a_U\,\mathrm{d}a_S}.$$

Employment in market j rises when its unemployed workers meet a vacancy and the meeting becomes a match, and falls when a shock redraws the quality of a match below the reservation quality or the worker leaves. Integrating (32) over types, and (34) over types and qualities, in which the flows between qualities cancel, gives

$$\kappa_U \iint \mathbf{1}\{S_U(\cdot,1) \geq 0\}\,(u_U + d\,u_S)\,\mathrm{d}a_U\,\mathrm{d}a_S = \iint \big(\lambda_U G(p^*) + \nu\big)\,e_U\,\mathrm{d}a_U\,\mathrm{d}a_S,$$
$$\kappa_S \iint (1-d)\big(1-\Gamma(p_S^*)\big)\,u_S\,\mathrm{d}a_U\,\mathrm{d}a_S = \iint \big(\lambda_S \Gamma(p_S^*) + \nu\big) \int e_S\,\mathrm{d}p\,\mathrm{d}a_U\,\mathrm{d}a_S.$$

Let ψ_j be the share of meetings between a vacancy and an unemployed worker of market j that become a match, and ω_j the share of quality shocks that end a match of market j. Dividing by the labour force of market j gives

$$\kappa_j\,\psi_j\,\bar{u}_j = (\lambda_j\,\omega_j + \nu)(1 - \bar{u}_j), \tag{36}$$

and substituting $\kappa_j = \mu_j \sqrt{\mathcal{V}_j/\widetilde{\mathcal{U}}_j}$ gives the vacancies of market j as a function of its unemployment rate:

$$\mathcal{V}_j = \widetilde{\mathcal{U}}_j \left[\frac{(\lambda_j\,\omega_j + \nu)(1-\bar{u}_j)}{\mu_j\,\psi_j\,\bar{u}_j}\right]^2. \tag{37}$$

The job seekers $\widetilde{\mathcal{U}}_j$ are the unemployed workers of market j and, in the skilled market, also the workers who search on the job. At given ψ_j and ω_j, a given labour force and given on-the-job searchers, (37) gives vacancies as a decreasing function of the unemployment rate: this is the Beveridge curve of market j. In a stationary equilibrium, the tightness pair sets the point on each curve.

A change in tightness alone moves market j along its curve. A change in μ_j, λ_j or ν shifts it, and so does a change in ψ_j, ω_j, the labour force or the on-the-job searchers, which depend on the reservation qualities and on the decisions τ, d and s: these set who is unemployed, who works and who searches in each market. Through the values, every parameter, and the tightness pair itself, moves the reservation qualities and the decisions.

6 Transition

A transition is the path the economy follows from one stationary equilibrium to another when the parameters change. The economy starts in the stationary equilibrium at the initial parameters, and no agent expects the change. I measure time t from the change, so the parameters take their new values from $t = 0$ on. From then on agents know the new parameters and the path the economy follows, and every choice is optimal given that path. At every t the economy is in equilibrium: given the path, no worker or firm wants to act differently, and free entry holds. It is not stationary, because the densities are away from those of the new stationary equilibrium, and tightness and the value functions change over time as the densities move. Heckman et al. (1998) compute such a path for a model of schooling and skill formation, and Eeckhout and Lindenlaub (2019) for a frictional labour market with on-the-job search.

When the parameters change, the densities stay where they are, but the value functions respond at once. At the new parameters, training, search in each market and each match are worth different amounts than in the initial equilibrium, so workers train, search and keep matches differently, and firms open a different number of vacancies. These choices change the flows into and out of every state, and the flows move the densities towards the stationary equilibrium at the new parameters. Part of the move happens at once: matches no longer worth keeping end, and the unskilled unemployed of the types that now train start training. The rest takes time, at the speed of the flows. Unemployment and matches adjust at the rates at which workers find and lose jobs. The trained population adjusts only as trainees complete and workers are replaced, at rates ϕ and ν.

6.1 Value functions

Every object below is a function of $t \geq 0$, and a dot denotes a derivative with respect to t. Each value function satisfies its steady state equation plus its change over time. Each decision is taken at every t. Only the value functions that change in a further way are shown here: the values of search and the surpluses. Appendix A writes out the rest.

Unskilled unemployed workers. A worker who searches now can still train later, so the value she continues with is U_U, not the value of search:

$$(r+\nu)\,U_U^{(0)}(\mathbf{a},t) = b_U + \kappa_U(t)\big[E_U(\mathbf{a},1,t) - U_U(\mathbf{a},t)\big]^+ + \dot{U}_U(\mathbf{a},t). \tag{38}$$

This is why the value of search depends on a_S along the transition.

Skilled unemployed workers. A worker who searches one market now can change market later, so the value she continues with in either market is U_S:

$$(r+\nu)\,U_S^{(0)}(\mathbf{a},t) = b_S + \kappa_S(t)\int \big[E_S(\mathbf{a},p,t) - U_S(\mathbf{a},t)\big]^+ \,\mathrm{d}\Gamma(p) + \dot{U}_S(\mathbf{a},t), \tag{39}$$

$$(r+\nu)\,U_S^{(1)}(\mathbf{a},t) = b_S + \kappa_U(t)\big[E_U(\mathbf{a},1,t) - U_S(\mathbf{a},t)\big]^+ + \dot{U}_S(\mathbf{a},t). \tag{40}$$

Matches. The unskilled surplus gains its own change over time and the change in the worker's value of unemployment:

$$(r+\nu+\lambda_U)\,S_U(\mathbf{a},p,t) = A\,P_U\,a_U\,p - \big[(r+\nu)\,U_U(\mathbf{a},t) - \dot{U}_U(\mathbf{a},t)\big] + \lambda_U\int_{p^*(\mathbf{a},t)}^1 S_U(\mathbf{a},p',t)\,\mathrm{d}G(p') + \dot{S}_U(\mathbf{a},p,t). \tag{41}$$

By staying in the match the worker gives up the return on U_U, but not its change: she can leave at any t and collect U_U at its value then. The two skilled surpluses gain the same two terms, with U_S in place of U_U. Wages are bargained again whenever the value functions change, so the sharing rule (6) holds at every t, and a match ends as soon as its surplus is negative.

Vacant jobs. Firms open and close vacancies without delay, so $V_j(t) = 0$ at every t, and the free-entry condition (35) holds at every t:

$$\frac{\kappa_j(t)}{\theta_j(t)}\,\mathcal{J}_j(t) = k_j, \qquad j \in \{U,S\}. \tag{42}$$

Tightness is not a state variable: it is set at each t by the value functions and the job seekers at t.

6.2 Laws of motion

Along the transition the flows of Section 4.1 need not balance, and each density changes by its inflows less its outflows at the rates and decisions at t. A match also ends when a rising reservation quality passes its quality, so unskilled matches are distributed over quality, as skilled matches are: $e_U(\cdot,p)$ is the density of unskilled matches of quality $p < 1$. Every density and decision is a function of the type and of t, and I drop the arguments. Only the densities of trainees and of skilled unemployed workers, where the education margin acts, are shown here; the others change in the same way.

Newborns and workers whose unskilled match ends enter unskilled unemployment, and those of a type that trains start training at once. Trainees leave by completing, at rate ϕ, or by leaving the economy, at rate ν:

$$\dot{t} = \tau\left[\nu\,\ell + \lambda_U G(p^*)\,e_U + \max\{\dot{p}^*,0\}\,e_U(\cdot,p^*)\right] - (\phi+\nu)\,t. \tag{43}$$

Skilled unemployed workers arrive when trainees complete or a skilled match ends, and leave when they find a feasible match in the market they search or leave the economy:

$$\dot{u}_S = \phi\,t + \lambda_S\Gamma(p_S^*)\int e_S(\cdot,p)\,\mathrm{d}p + \max\{\dot{p}_S^*,0\}\,e_S(\cdot,p_S^*) - \Big((1-d)\,\kappa_S\big(1-\Gamma(p_S^*)\big) + d\,\kappa_U\mathbf{1}\{S_U(\cdot,1)\geq 0\} + \nu\Big)\,u_S. \tag{44}$$

The term $d\,\kappa_U \mathbf{1}\{S_U(\cdot,1) \geq 0\}\, u_S$ is the cross-market flow: skilled workers who accept an unskilled job, which ends their skilled status. It can be positive on the path even where it is zero in both stationary equilibria, because a worker who trained under the initial parameters may find searching the unskilled market worth more than searching the skilled market under the new parameters.

6.3 Transition equilibrium

A transition equilibrium is a path, for $t \geq 0$, of a tightness pair (θ_U, θ_S), reservation qualities p^* and p_S^*, decisions τ, d and s, and densities that satisfy at every t the free-entry conditions (42), the job destruction conditions (14) and (27), the decision rules of Section 6.1 and the laws of motion of Section 6.2, starting from the densities of the initial stationary equilibrium and converging to a stationary equilibrium at the new parameters.

Appendix A The transition

Unemployed workers. The choices (7) and (16) are made at every t, with the values of search (38)–(40). The value of training gains its change over time:

$$(r + \phi + \nu)\, T = \phi\, U_S + \dot{T}. \tag{45}$$

Matches. The values of a match to the worker and to the firm, (10)–(11) in the unskilled market and (19)–(22) in the skilled market, each gain their change over time. Adding the two cancels the wage, and since $E_j + J_j = S_j + U_j$ at every quality and every t, subtracting the worker's value of unemployment gives (41) and its two skilled counterparts. The reservation qualities are (14) and (27) at every t, and a skilled worker searches on the job if and only if $S_S^{(1)} \geq S_S^{(0)}$.

Vacant jobs. The values of a vacancy (28) and (29) gain their change over time, and $\mathcal{J}_j$ is computed from the value functions and densities at t. With $V_j = \dot{V}_j = 0$ at every t, they give (42).

Appendix B Computing the equilibria

B.1 A stationary equilibrium

The equilibrium is characterised by the tightness pair, the value functions and surpluses, the decisions and the densities. Given (θ_U, θ_S), which are among the estimated parameters, each step below uses only what the previous ones deliver:

1. At $\kappa_U = \mu_U \sqrt{\theta_U}$, solve (9), (12) and (14) for the value of unskilled search, the unskilled surplus and p^*.
2. At $\kappa_S = \mu_S \sqrt{\theta_S}$, and with $E_U(\cdot,1)$ from step 1, solve (17)–(18) and (23)–(24) for U_S, the skilled surpluses, s, p_S^* and d.
3. Compute T from (8) and τ from (7).

4. Given every decision, the flow conditions of Section 4.1 are linear in the densities; solve them type by type.
5. Compute k_U and k_S from (35).

Steps 1 and 2 are fixed points in the value functions, which I solve by iteration. At given vacancy costs, I solve for the tightness pair instead: from a guess, I apply steps 1 to 4 and move each θ_j towards $(\mu_j \mathcal{J}_j / k_j)^2$, the tightness at which (35) holds at the current value functions and densities, until free entry holds.

B.2 A transition

The transition is characterised by the two tightness paths. Starting from paths constant at the tightness pair of the new stationary equilibrium, I iterate on three steps until free entry holds at every date:

1. Solve the value functions and decisions backward, from a distant date at which they are those of the new stationary equilibrium.
2. Advance the densities forward from those of the initial stationary equilibrium with the laws of motion of Section 6.2.
3. At each date, move the tightness pair towards the one at which (42) holds given that date's value functions and densities.

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